GEK-83870

DRIVE SYSTEM INSTRUCTIONS

DS3828 SILTRON™ Power Conversion Module Troubleshooting Guide for Metal Industry Applications

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DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

GENERAL ELECTRIC

DS3828 SILTRON™ Power Conversion Module Troubleshooting Guide for Metal Industry Applications

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These instructions do not purport to cover all details or variations in equipment, nor to provide for every possible contingency to be met during installation, operation, and maintenance. Should further information be desired or should particular problems arise that are not covered sufficiently for the purchaser's purpose, the matter should be referred to General Electric Company, Drive Systems Operations, Salem, Virginia, U.S.A.

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Safety Symbol Legend

WARNING Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in personal injury or death.

CAUTION Commands attention to an operating procedure, practice, condition or statement, which, if not strictly observed, could result in damage to or destruction of equipment.

WARNING

THIS EQUIPMENT CONTAINS A POTENTIAL HAZARD OF ELECTRICAL SHOCK OR BURN. ONLY PERSONNEL WHO ARE ADEQUATELY TRAINED AND THOROUGHLY FAMILIAR WITH THE EQUIPMENT AND THESE INSTRUCTIONS SHOULD INSTALL, OPERATE OR MAINTAIN THIS EQUIPMENT.

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

TO MINIMIZE HAZARD OF ELECTRICAL SHOCK OR BURN, APPROVED GROUNDING PRACTICES AND PROCEDURES MUST BE FOLLOWED.

INTRODUCTION

This manual is a guide to the troubleshooting of General Electric Company's DS3828 SILTROM℠ Power Conversion Module for metal industry applications. This manual contains information, including diagrams and flowcharts, to help you find and correct problems that might occur. This manual is used in conjunction with another GE publication, the ***DS3828 SILTRON℠ Power Conversion Module Maintenance Manual for Metal Industry Applications.***

We assume that a user of this manual has knowledge and experience in electrical and electronics technology, as well as prior experience with industrial control systems.

In order to properly maintain the SILTRON power conversion equipment, you need the following additional documentation:

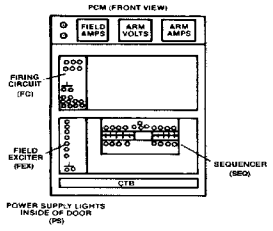
- System elementary diagram
- DS3828 SILTRON℠ Power Conversion Module Maintenance Manual for Metal Industry Applications (GEK-83806)

You also need the following equipment:

- Multimeter or digital voltmeter
- Hand tachometer
- Strip chart recorder
- Dual-trace oscilloscope

INSTRUCTIONS

HEAVY LINED BOXES ARE 'MILESTONE' CHECKS
EX: WHEN DRIVE WILL NOT RUN AND 'M' PICKS UP PROBLEM IS BELOW THAT MILESTONE BOX
HEAVY LINES MEANS COMMON FLOW.USE TO ELIMINATE POSSIBLE PROBLEM AREAS.
EX: WHEN DRIVE WILL NOT RUN, PUSH JOG FIRST: IF THIS WORKS ALL COMMON ELEMENTS ARE OK



DRIVE WILL NOT MOVE

PS — IS YELLOW LIGHT ON PWR SUP ON — NO → 5V PWR SUP PROB. SEE SUBROUTINE 1 OR 1A → SEE SH. 4

YES

SEQ IMOK — IS YELLOW IMOK LIGHT ON SEQ BD ON — NO → REPLACE SEQUENCER BOARD

YES

SEQ ORST

FC IMOK — IS YELLOW IMOK LIGHT ON FIR CKT BD ON — NO → IS ORST LIGHT OFF — NO

YES → REPLACE SEQ BOARD OR FIR CKT BD

YES

FAULT

SEQ F — IS FAULT LIGHT OFF — NO → SEE SH. 3

YES

OUT — IS JUMPER J106 IN – OUT — IN → IS CEMF SEL LIGHT ON — NO → REPLACE SEQ BOARD

YES

SEQ CEMF SEL.

SEQ RR — IS READY TO RUN LIGHT ON — NO → ARE ANY OF THE INPUT LIGHTS ON — YES → IS QUICK STOP LIGHT ON (SEQ. QSR) — NO → IS ESTP LIGHT ON (SEQ. ESTP) — YES → IS OFF LIGHT ON (SEQ. OFF) — YES → IS ON LIGHT ON (SEQ ON) — YES → REPLACE SEQ BOARD

YES

NO → POSSIBLY105V POWER SUPPLY FAILURE SEE SUBROUTINE 1 → SEE SH. 4

YES → IS RUN PERMISSIVE LIGHT ON (SEQ. RPER) — NO → INPUT PROBLEM SEE SUB-ROUTINE 2

YES → IS J145 IN OR OUT — IN → VERIFY THAT JUMPER IS MAKING A GOOD CONTACT → REPLACE SEQ BOARD

OUT

NO → IS DRIVE STATIONARY — NO → ALLOW DRIVE TO STOP

YES → VERIFY ZERO SPD CONDITION ON CTB 1. POSSIBLE BAD NMEC ON CTB 1

RUN

OUT — JUMPER J6S IN – OUT

IN

SEQ AUTO — IS AUTO LIGHT ON — NO → HAS AUTO MODE BEEN SELECTED — YES → INPUT PROB SEE SUB-ROUTINE 2

YES

NO → SELECT AUTO MODE

INITIATE A 'RUN' CONDITION

SEQ RUN — IS RUN LIGHT ON — NO → INPUT PROB SEE SUB-ROUTINE 2

YES

MANUAL

OUT — JUMPER J5S IN – OUT

IN

SEQ MAN — IS MAN LIGHT ON — NO → HAS MAN MODE BEEN SELECTED — YES → INPUT PROB SEE SUB-ROUTINE 2 → SEE SH. 5

YES

NO → SELECT MAN MODE

FORWARD — **REVERSE**

INITIATE A FORWARD CONDITION — INITIATE A REVERSE CONDITION

CURRENT MODE

INITIATED A CURR MODE CONDITION

SEQ CURM — IS CURM LIGHT ON — YES

NO → INPUT PROB SEE SUB-ROUTINE 2

SEQ FWD — IS FWD LIGHT ON — NO → INPUT PROB SEE SUB-ROUTINE 2

YES

SEQ REV — IS REV LIGHT ON — NO → INPUT PROB SEE SUB-ROUTINE 2

YES

SEQ CLM

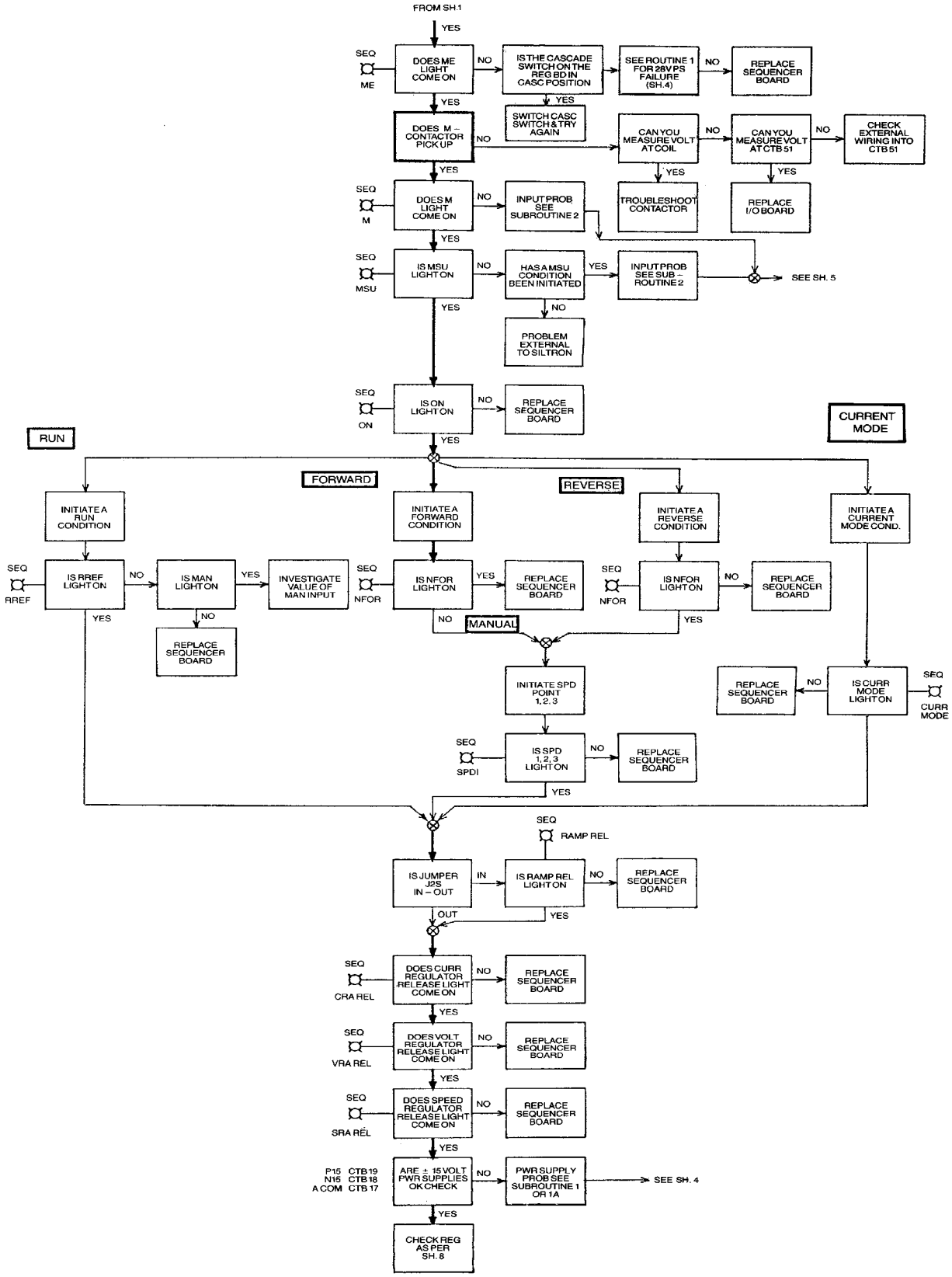
IS JUMPER J4S IN – OUT — OUT → IS JUMPER J9S IN – OUT — IN → IS CLM LIGHT ON — NO → HAS A CLM CONDITION BEEN INITIATED — YES → INPUT PROB SEE SUB-ROUTINE 2

IN — OUT — YES

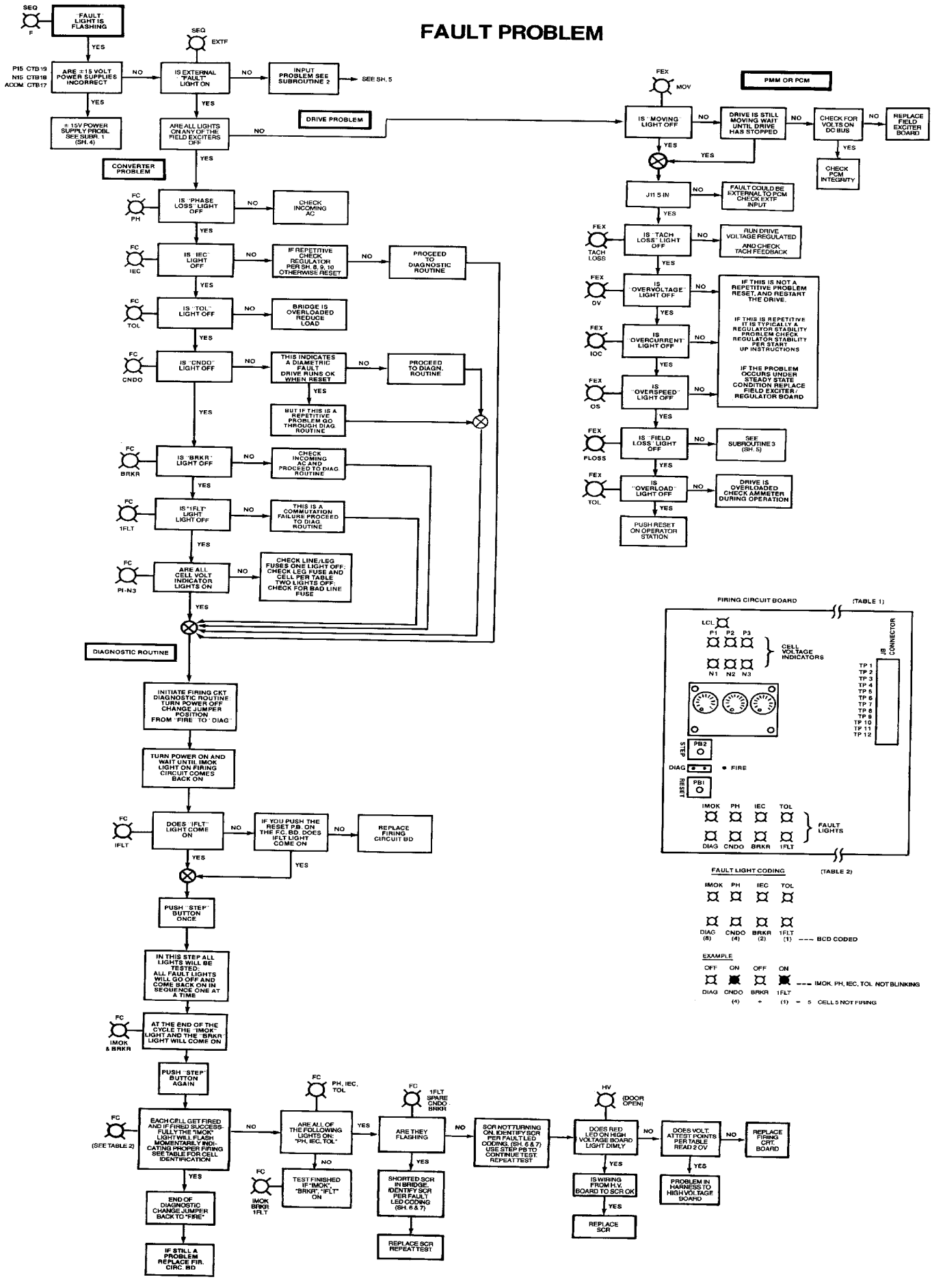
NO → PROBLEM EXTERNAL TO SILTRON

CONT ON SH. 2

SEE SH. 5


FROM SH.1
YES
SEQ
ME
DOES ME LIGHT COME ON
NO
IS THE CASCADE SWITCH ON THE REG BD IN CASC POSITION
SEE ROUTINE 1 FOR 28V PS FAILURE (SH.4)
NO
REPLACE SEQUENCER BOARD
YES
YES
SWITCH CASC SWITCH & TRY AGAIN
DOES M – CONTACTOR PICK UP
NO
CAN YOU MEASURE VOLT AT COIL
NO
CAN YOU MEASURE VOLT AT CTB 51
NO
CHECK EXTERNAL WIRING INTO CTB 51
YES
YES
SEQ
M
DOES M LIGHT COME ON
NO
INPUT PROB SEE SUBROUTINE 2
TROUBLESHOOT CONTACTOR
REPLACE I/O BOARD
YES
SEQ
MSU
IS MSU LIGHT ON
NO
HAS A MSU CONDITION BEEN INITIATED
YES
INPUT PROB SEE SUB – ROUTINE 2
SEE SH. 5
YES
NO
PROBLEM EXTERNAL TO SILTRON
SEQ
ON
IS ON LIGHT ON
NO
REPLACE SEQUENCER BOARD
YES
CURRENT MODE
RUN
FORWARD
REVERSE
INITIATE A RUN CONDITION
INITIATE A FORWARD CONDITION
INITIATE A REVERSE CONDITION
INITIATE A CURRENT MODE COND.
SEQ
RREF
IS RREF LIGHT ON
NO
IS MAN LIGHT ON
YES
INVESTIGATE VALUE OF MAN INPUT
YES
NO
REPLACE SEQUENCER BOARD
SEQ
NFOR
IS NFOR LIGHT ON
YES
REPLACE SEQUENCER BOARD
NO
SEQ
NFOR
IS NFOR LIGHT ON
NO
REPLACE SEQUENCER BOARD
YES
MANUAL
INITIATE SPD POINT 1, 2, 3
REPLACE SEQUENCER BOARD
NO
IS CURR MODE LIGHT ON
SEQ
CURR MODE
SEQ
SPDI
IS SPD 1, 2, 3 LIGHT ON
NO
REPLACE SEQUENCER BOARD
YES
SEQ
RAMP REL
IS JUMPER J2S IN – OUT
IN
IS RAMP REL LIGHT ON
NO
REPLACE SEQUENCER BOARD
OUT
YES
SEQ
CRA REL
DOES CURR REGULATOR RELEASE LIGHT COME ON
NO
REPLACE SEQUENCER BOARD
YES
SEQ
VRA REL
DOES VOLT REGULATOR RELEASE LIGHT COME ON
NO
REPLACE SEQUENCER BOARD
YES
SEQ
SRA REL
DOES SPEED REGULATOR RELEASE LIGHT COME ON
NO
REPLACE SEQUENCER BOARD
YES
P15 CTB 19
N15 CTB 18
A COM CTB 17
ARE ± 15 VOLT PWR SUPPLIES OK CHECK
NO
PWR SUPPLY PROB SEE SUBROUTINE 1 OR 1A
SEE SH. 4
YES
CHECK REG AS PER SH. 8

FAULT PROBLEM



SUBROUTINE 1 (FOR NPSE)

POWER SUPPLY PROBLEM: 5 VOLT, +15 VOLT, −15 VOLT, +28 VOLT, +105 VOLT, & 115VAC

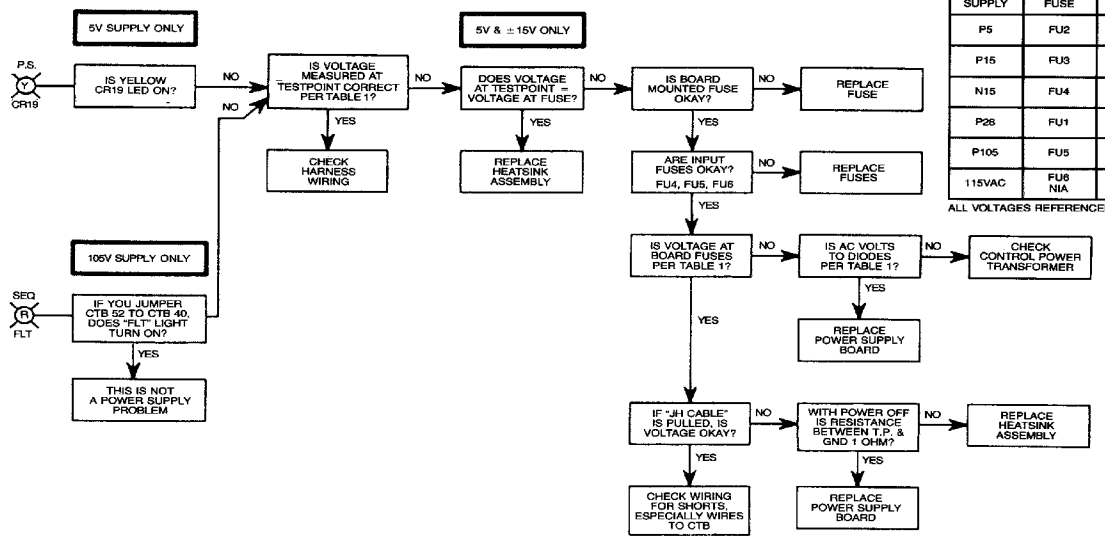


TABLE 1

POWER SUPPLY	VOLTS AT FUSE	TEST POINT	VOLTS AT DIODE
P5	FU2	+5V ON P.S.	8V AT CR11-16
P15	FU3	+15V ON P.S.	17.5V AT CR21-26
N15	FU4	−15V ON P.S.	17.5V AT CR21-26
P28	FU1	+28V ON P.S.	20V AT CR1-6
P105	FU5	CTB 52 105V	NA
115VAC	FU8 NIA	NA	NA

ALL VOLTAGES REFERENCED TO ACOM

SUBROUTINE 1A (FOR NPSS, DPSS)

POWER SUPPLY PROBLEM: 5 VOLT, +15 VOLT, −15 VOLT, +28 VOLT, +105 VOLT, & 115VAC

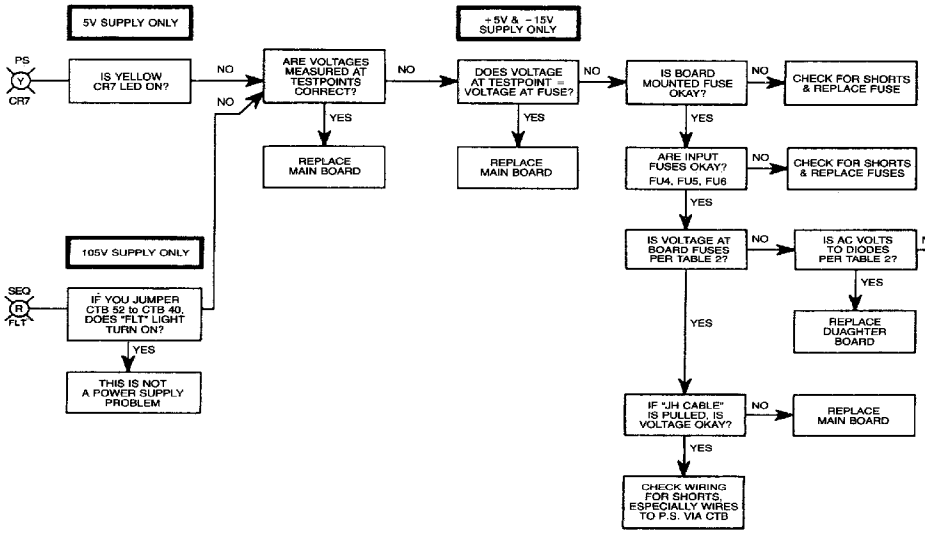


TABLE 2

POWER SUPPLY	VOLTS AT FUSE	TEST POINT	VOLTS AT DIODE
P5	17VDC FU2	+5V	8VAC AT CR19-30
P15	22VDC FU3	+15V	17.5VAC AT CR7-12
N15	−22VDC FU4	−15V	17.5VAC AT CR13-18
P28	+28VDC FU1	+28V	20VAC AT CR1-6
P105	−105VDC FU5	+105V	115VAC AT CR31-34
115VAC	+115VAC FU6	NA	NA

ALL VOLTAGES REFERENCED TO ACOM

SUBROUTINE 2

INPUT PROBLEM

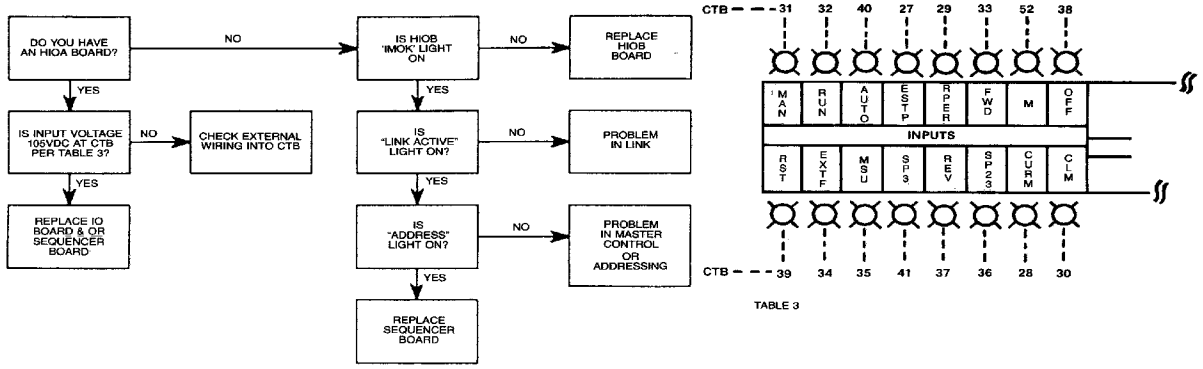
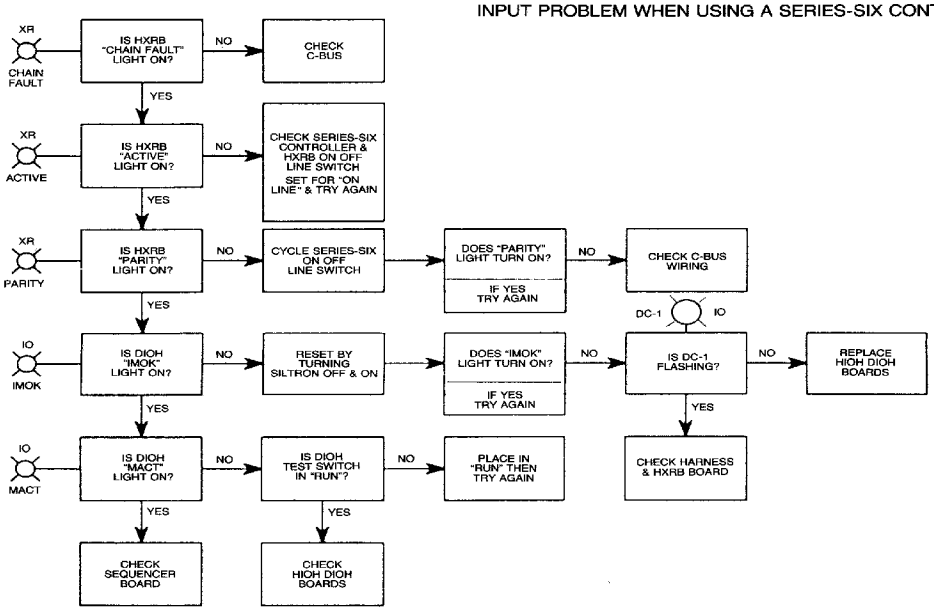


TABLE 3

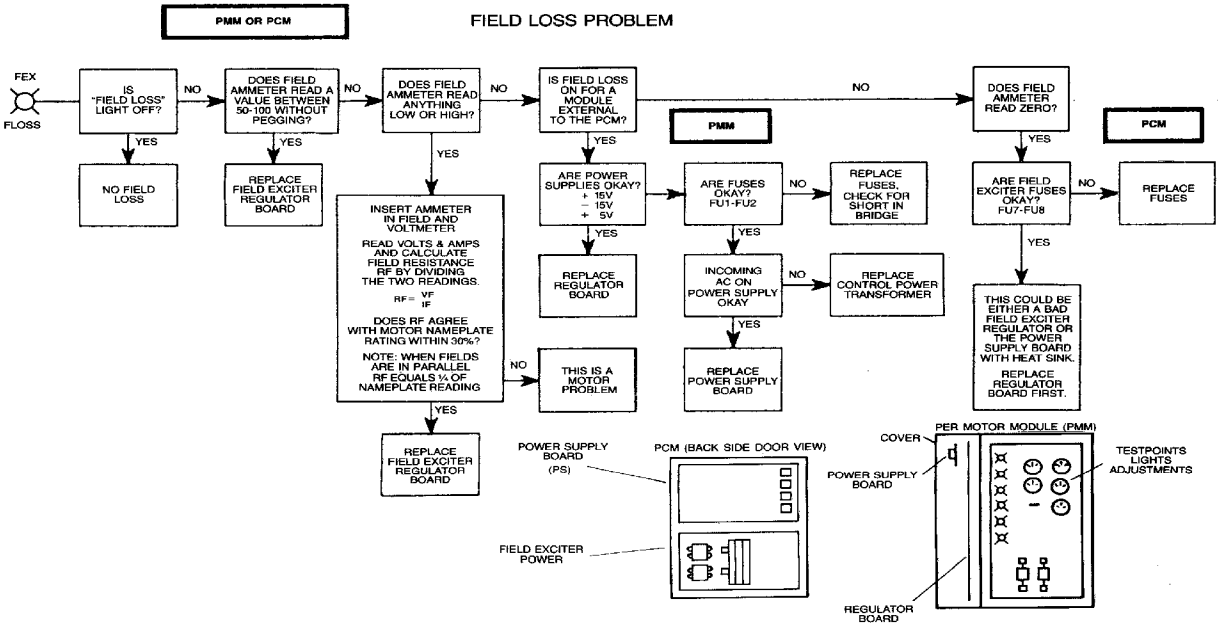
SUBROUTINE 2A

INPUT PROBLEM WHEN USING A SERIES-SIX CONTROLLER



SUBROUTINE 3

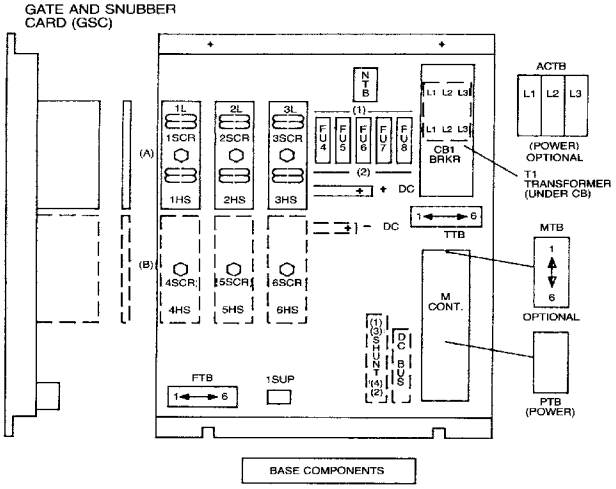
FIELD LOSS PROBLEM



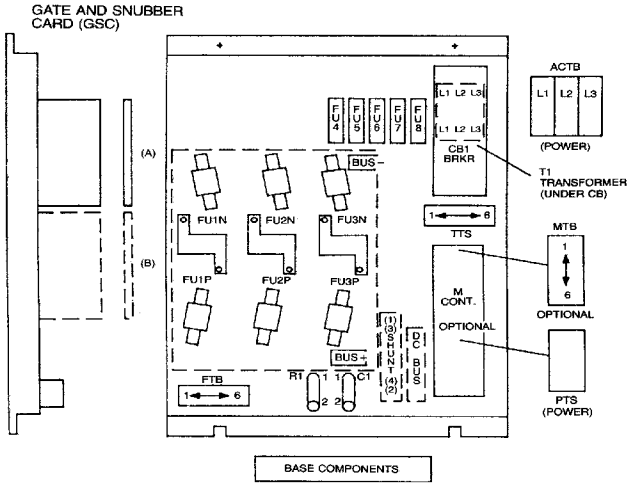
CELL FUSE IDENTIFICATION

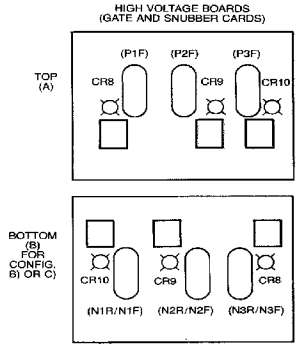
LOW HORSEPOWER

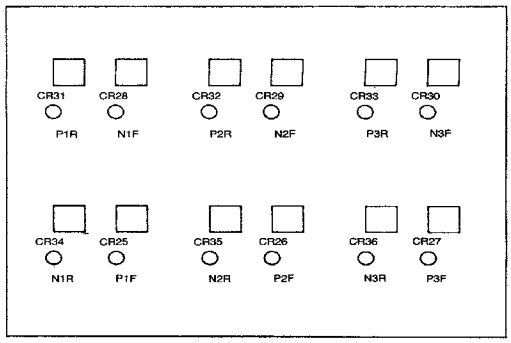
6 PULSE NON-REVERSING



6 PULSE REVERSING

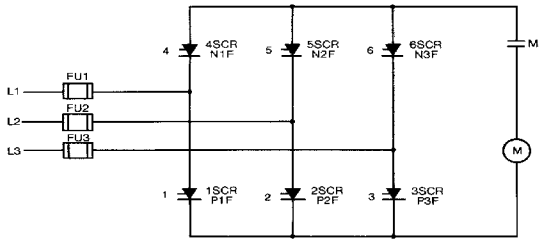




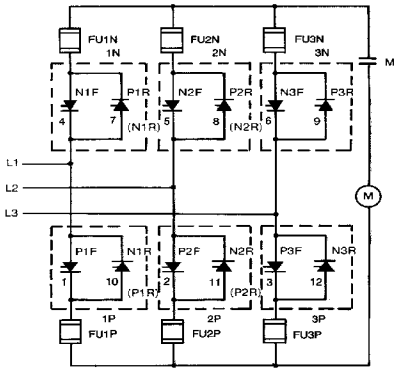


CONFIGURATIONS

A.) 6-ULSE NON-REVERSING



B.) 6-PULSE REVERSING

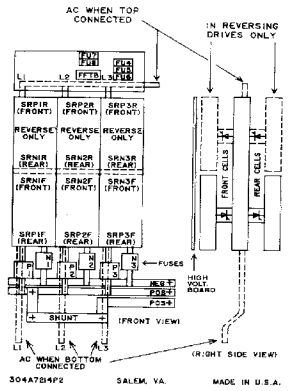


CELL	NOMEN-CLATURE	CONFIG.	FUSE FOR CONFIG.	LOW
1	SRP1F	A,B	1P	CR25
2	SRP2F	A,B	2P	CR26
3	SRP3F	A,B	3P	CR27
4	SRN1F	A,B	1N	CR28
5	SRN2F	A,B	2N	CR29
6	SRN3F	A,B	3N	CR30
7	SRP1R	B	1N	CR31
8	SRP2R	B	2N	CR32
9	SRP3R	B	3N	CR33
10	SRN1R	B	1N	CR34
11	SRN2R	B	2N	CR35
12	SRN3R	B	3N	CR36

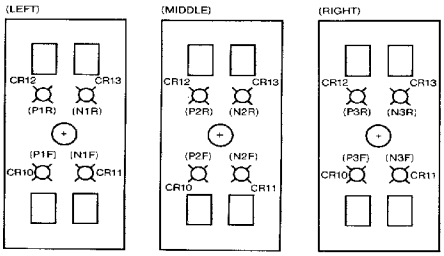
EXAMPLES

1) CELL 4 SHORTED – CHECK CELL 4 (N1F AND/OR FUSE 1) IF CONFIGURATION "A". CHECK CELLBLOCK 1N AND/OR FUSE FU1N IF CONFIGURATION "B".
2) CELL 5 NOT FIRING – CHECK CELL 5 (N2F) IF 1N CONFIG. "A". CHECK CELLBLOCK 2N IF 1N CONFIG. "B".

MEDIUM HORSEPOWER



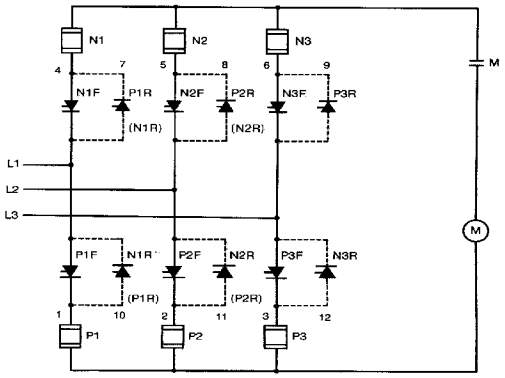
HIGH VOLTAGE BOARDS (3)



EXAMPLES:
1) CELL 4 SHORTED – CHECK CELL 4 AND 7 (SRN1F/SRP1R) AND/OR FUSE N1
2) CELL 5 NOT FIRING – CHECK CELL 5 (SRN2F) AND OR FUSE N2

CONFIGURATIONS

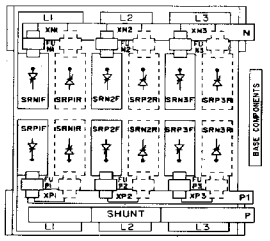
DOTTED LINE FOR REVERSING
a) NON REV. 6P
b) REV. 6P



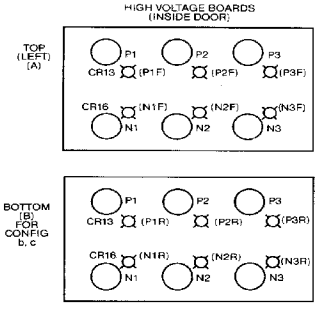
CELL	NOMEN-CLATURE	CONFIGU-RATION	FUSE	HIGH VOLTAGE BOARD MEDIUM HP
1	SRP1F	a, b	P1	CR10(L)*
2	SRP2F	a, b	P2	CR10(M)
3	SRP3F	a, b	P3	CR10(R)
4	SRN1F	a, b	N1	CR11(L)
5	SRN2F	a, b	N2	CR11(M)
6	SRN3F	a, b	N3	CR11(R)
7	SRP1R	b	P1	CR12(L)
8	SRP2R	b	P2	CR12(M)
9	SRP3R	b	P3	CR12(R)
10	SRN1R	b	N1	CR13(L)
11	SRN2R	b	N2	CR13(M)
12	SRN3R	b	N3	CR13(R)

* L – LEFT, M = MIDDLE, R = RIGHT

LARGE HORSEPOWER



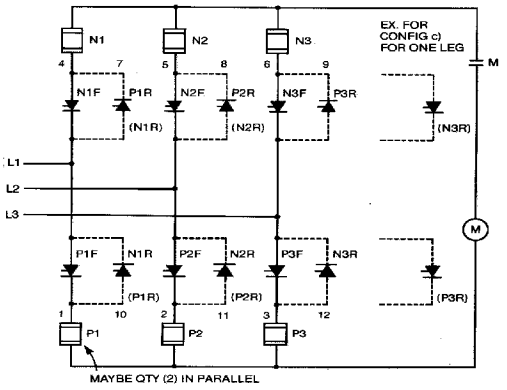
HIGH VOLTAGE BOARDS (INSIDE DOOR)



EXAMPLES:
SAME AS FOR MEDIUM HORSEPOWER

CONFIGURATIONS

DOTTED LINE FOR REVERSING OR "SUPERBRIDGE"
a) NON REV. 6P
b) REV. 6P
c) NON REV. 6P ("SUPERBRIDGE")



CELL	NOMEN-CLATURE	CONFIGU-RATION*	FUSE	HIGH VOLTAGE BOARD LARGE HP
1	SRP1F	a, b	P1	CR13(T)*
2	SRP2F	a, b	P2	CR14(T)
3	SRP3F	a, b	P3	CR15(T)
4	SRN1F	a, b	N1	CR16(T)
5	SRN2F	a, b	N2	CR17(T)
6	SRN3F	a, b	N3	CR18(T)
7	SRP1R	b, c	P1	CR13(B)
8	SRP2R	b, c	P2	CR14(B)
9	SRP3R	b, c	P3	CR15(B)
10	SRN1R	b, c	N1	CR16(B)
11	SRN2R	b, c	N2	CR17(B)
12	SRN3R	b, c	N3	CR18(B)

NOTE: IN CONFIGURATION c) "REVERSE" SCR's P1R-N3R ARE IN PARALLEL (ie. ANODE – ANODE) WITH "FORWARD" SCR's P1F-N3F FOR HIGHER AMPERAGE. NOMENCLATURE AS INDICATED IN PARENTHESIS ( )

ALSO PARALLEL CELLS (ie P1F/P1R) WILL BE FIRED SIMULTANEOUSLY

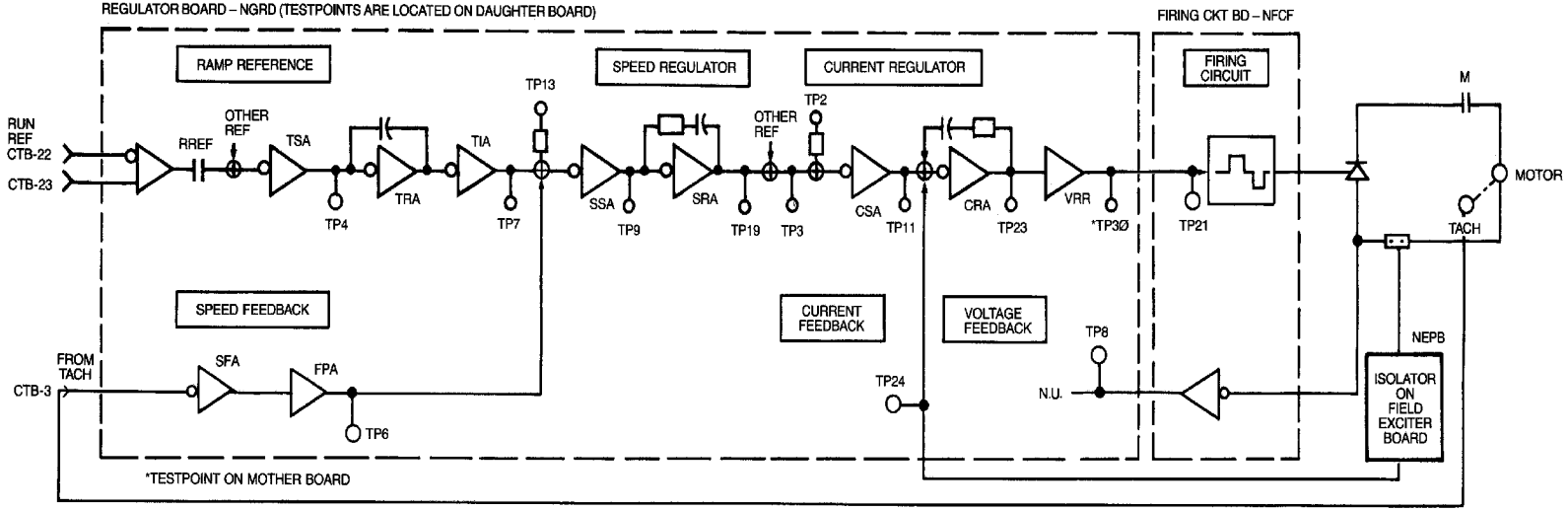
*T – TOP, B – BOTTOM

CASCADE TEST/TROUBLESHOOTING PROCEDURES

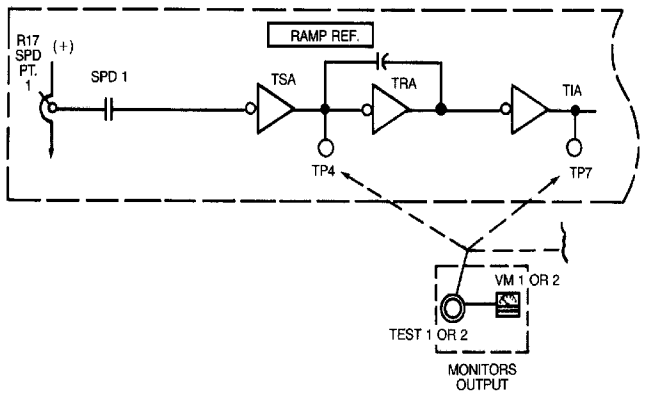
SIMPLIFIED BLOCK DIAGRAM OF SPD/CURR REGULATED DRIVE (J7R, J41R, J4R, J37R)

DESCRIPTION OF REGULATOR

THIS DRIVE REGULATOR CONSISTS OF A SPEED REGULATOR WITH AN INNER CURRENT LOOP. THE MAIN REFERENCE TO THE SPEED REGULATOR COMES FROM A RAMP REFERENCE CIRCUIT PROVIDING A LINEAR RAMP ADJUSTABLE FOR ACCELERATION AND DECELERATION. THE REFERENCE ITSELF MAY COME FROM A RHEOSTAT OR A MASTER CONTROL. ADDITIONAL REFERENCES SUCH AS TENSION SIGNAL PROVIDING AN OUTER LOOP. SPEED FEEDBACK IS GENERATED BY A TACHOMETER AND THEN SCALED BY THE TWO SPEED FEEDBACK AMPLIFIERS, CURRENT FEEDBACK IS FROM A SHUNT IN THE ARMATURE LOOP, ISOLATED, SCALED AND FED BACK VIA THE FIELD EXCITER, VOLTAGE FEEDBACK IS SENESED IN THE ARMATURE LOOP AND FED BACK VIA FIRING CIRCUIT BOARD.



CASCADE TEST/TROUBLE SHOOTING PROCEDURES



OBJECTIVES AND ASSUMPTIONS

THE OBJECTIVE OF THE FOLLOWING PROCEDURE IS TO DETERMINE QUICKLY AND WITHOUT EXTERNAL EQUIPMENT WHETHER THE PROBLEM IS RELATED TO THE DRIVE REGULATOR OR ANY EXTERNAL EQUIPMENT LIKE TACH, MOTOR, OR CONVERTER. IT IS ASSUMED THAT THIS DRIVE HAS RUN CORRECTLY PRIOR TO THE FAILURE AND NO JUMPERS OR ADJUSTMENTS HAVE BEEN CHANGED.

TEST PROCEDURE

WITH THE DRIVE STOPPED CHANGE THE CASCADE SWITCH MOUNTED ON THE REGULATOR DAUGHTER BOARD TO THE "CASCADE" POSITION. THIS WILL PREVENT THE "M" CONTACTOR FROM PICKING UP AND THEREBY PREVENTING THE DRIVE FROM RUNNING. ALSO ALL SUICIDES WILL BE RELEASED AND OPERATIONAL AMPLIFIERS IN SPEED, VOLT, AND CURRENT REGULATOR WILL BE CHANGED TO FIXED GAIN. ALSO THE SPEED POINT 1 FORWARD REFERENCE IS AUTOMATICALLY INJECTED.

WAIT UNTIL REFERENCE AT TP 7 HAS RAMPED UP COMPLETELY AND THEN START TRACING THE SIGNAL THROUGH THE REGULATOR AND COMPARE THEM WITH THE VOLTAGES LISTED IN THE FRONT SHEETS FOR THE PARTICULAR DRIVE. MAKE SURE RHEOSTAT R17 IS IN THE SAME POSITION AS IT WAS WHEN THESE NUMBERS WERE RECORDED. (REFER TO FRONT SHEETS OF ELEMENTARY.)

IF AN INCORRECT VOLTAGE LEVEL IS READ AT ANY TESTPOINT, MAKE SURE THAT NO OTHER REFERENCES COME INTO REGULATOR VIA FEEDBACKS OR OTHER INPUTS; THEN CHANGE REGULATOR BOARD.

CHANGE CASCADE SWITCH BACK TO "NORMAL" POSITION.

IF THE DRIVE IS STILL NOT RUNNING CORRECTLY, GO THROUGH NFCF DIAGNOSTIC ROUTINE AS EXPLAINED ON SHEET 3.

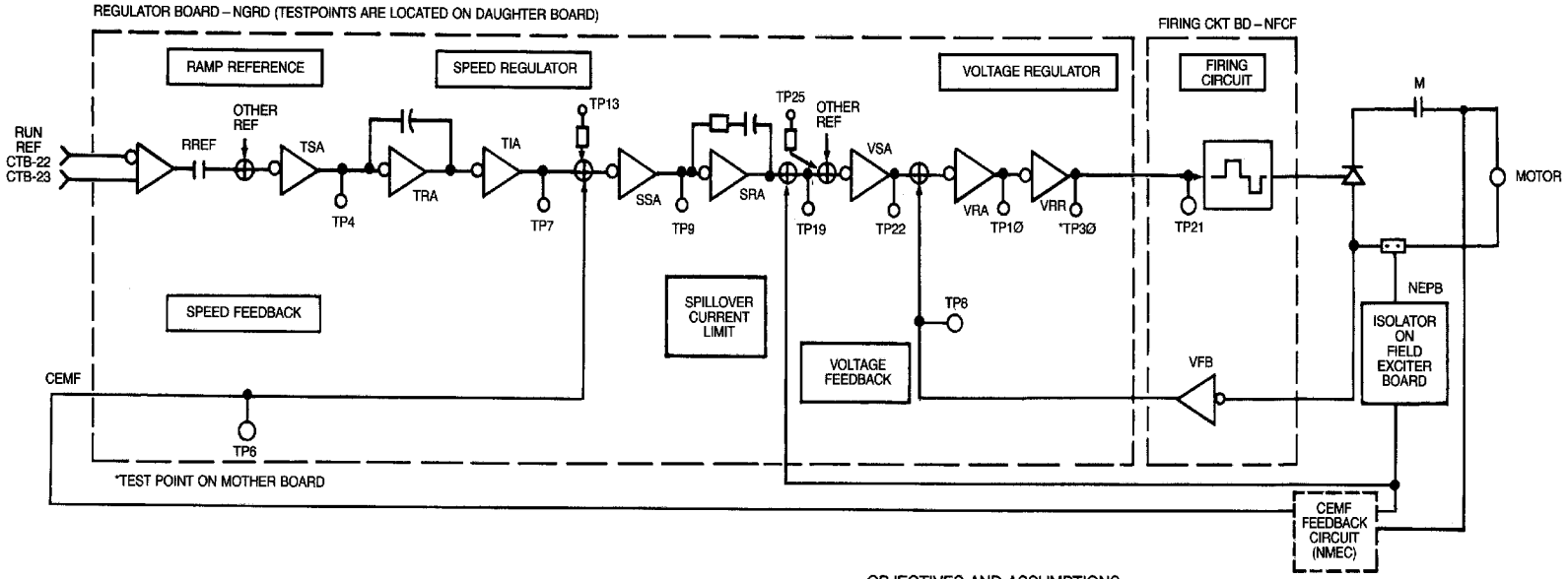
CASCADE TEST/TROUBLE SHOOTING PROCEDURES

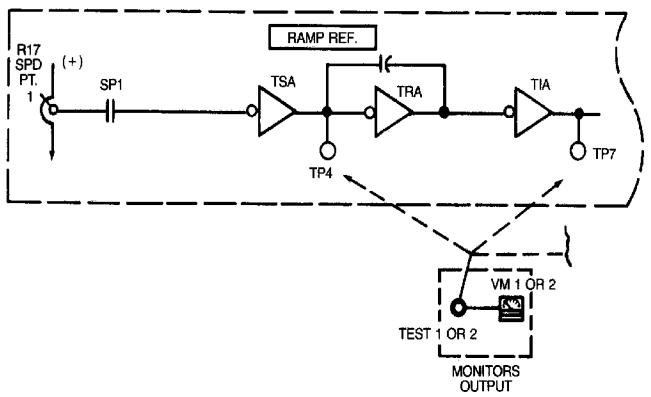
SIMPLIFIED BLOCK DIAGRAM OF CEMF/VOLT REGULATED DRIVE SPILLOVER CURRENT LIMIT (J7R, J4ØR, J36R)

DESCRIPTION OF REGULATOR

THIS DRIVE REGULATOR CONSISTS OF A CEMF REGULATOR WITH AN INNER VOLTAGE LOOP. THE MAIN REFERENCE TO THE SPEED REGULATOR COMES FROM A RAMP REFERENCE CIRCUIT PROVIDING A LINEAR RAMP ADJUSTABLE FOR ACCELERATION AND DECELERATION. THE REFERENCE ITSELF MAY COME FROM A RHEOSTAT OR A MASTER CONTROL. ADDITIONAL REFERENCES SUCH AS TENSION SIGNAL PROVIDING AN OUTER LOOP.

SPILLOVER CURRENT LIMIT IS FROM A SHUNT IN THE ARMATURE LOOP, ISOLATED, SCALED AND FED BACK VIA THE FIELD EXCITER, VOLTAGE FEEDBACK IS SENSED IN THE ARMATURE LOOP AND FED BACK VIA FIRING CIRCUIT BOARD.





OBJECTIVES AND ASSUMPTIONS

THE OBJECTIVE OF THE FOLLOWING PROCEDURE IS TO DETERMINE QUICKLY AND WITHOUT EXTERNAL EQUIPMENT WHETHER THE PROBLEM IS RELATED TO THE DRIVE REGULATOR OR ANY EXTERNAL EQUIPMENT LIKE TACH MOTOR, OR CONVERTER. IT IS ASSUMED THAT THIS DRIVE HAS RUN CORRECTLY PRIOR TO THE FAILURE AND NO JUMPERS OR ADJUSTMENTS HAVE BEEN CHANGED.

TEST PROCEDURE

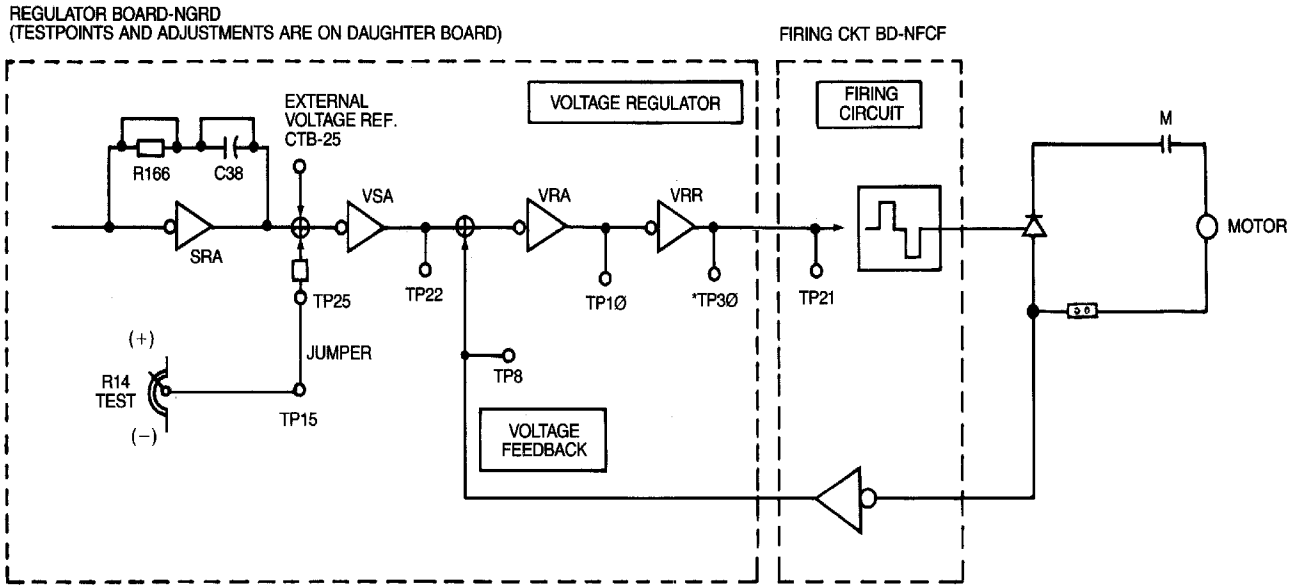
WITH THE DRIVE STOPPED CHANGE THE CASCADE SWITCH MOUNTED ON THE REGULATOR DAUGHTER BOARD TO THE "CASCADE" POSITION. THIS WILL PREVENT THE "M" CONTACTOR FROM PICKING UP AND THEREBY PREVENTING THE DRIVE FROM RUNNING. ALSO ALL SUICIDES WILL BE RELEASED AND OPERATIONAL AMPLIFIERS IN SPEED, VOLT, AND CURRENT REGULATOR WILL BE CHANGED TO FIXED GAIN. ALSO THE SPEED POINT 1 FORWARD REFERENCE IS AUTOMATICALLY INJECTED. WAIT UNTIL REFERENCE AT TP7 HAS RAMPED UP COMPLETELY AND THEN START TRACING THE SIGNAL THROUGH THE REGULATOR AND COMPARE THEM WITH THE VOLTAGES LISTED IN THE FRONT SHEETS FOR THE PARTICULAR DRIVE. MAKE SURE RHEOSTAT R17 IS IN THE SAME POSITION AS IT WAS WHEN THESE NUMBERS WERE RECORDED (REFER TO FRONT SHEETS OF ELEMENTARY)

IF AN INCORRECT VOLTAGE LEVEL IS READ AT ANY TESTPOINT, MAKE SURE THAT NO OTHER REFERENCES COME INTO REGULATOR VIA FEEDBACKS OR OTHER INPUTS; THEN CHANGE REGULATOR BOARD.

CHANGE CASCADE SWITCH BACK TO "NORMAL" POSITION.

IF THE DRIVE IS STILL NOT RUNNING CORRECTLY, GO THROUGH NFCF DIAGNOSTIC ROUTINE AS EXPLAINED ON SHEET 3.

RUNNING DRIVE VOLTAGE REGULATED FOR TEST PURPOSES



* TESTPOINT ON MOTHERBOARD

VERY OFTEN IT IS DESIRABLE TO RUN THE MOTOR OPEN LOOP EITHER FOR TEST PURPOSES OR DURING TROUBLESHOOTING. IT IS SOMETIMES VERY HARD TO JUDGE THE SOURCE OF THE PROBLEM, BECAUSE THE REGULATOR CAN EITHER CAUSE THE PROBLEM OR TRY TO CORRECT FOR IT. BY OPENING THE LOOP YOU CAN DETERMINE WHETHER YOUR PROBLEM IS IN THE DRIVE REGULATOR, POWER CONVERTER OR MOTOR.

FOLLOWING IS A QUICK AND EASY ROUTINE TO DO THIS

JUMPER J40R IN. THIS DIRECTS THE OUTPUT OF THE VOLTAGE REGULATOR TO THE FIRING CIRCUIT. SHORT R166 AND C38. THIS SUICIDES THE SPEED REGULATOR.

IF A REFERENCE COMES INTO CTB25, DISCONNECT THIS INPUT.

JUMPER TESTPOINTS TP15 TO TP25 THIS INJECTS A VARIABLE REFERENCE INTO THE VOLTAGE REGULATOR. SET R14 FOR ZERO VOLTS AT TP15.

START THE DRIVE THRU THE SEQUENCER USING THE NORMAL START/RUN PUSHBUTTON WHICH WILL RELEASE THE REGULATOR AND PICK UP THE DRIVE CONTACTOR.

SLOWLY TURN THE TEST POT R14 CLOCKWISE, WHICH WILL INCREASE ARMATURE VOLTAGE; COUNTERCLOCKWISE WILL DECREASE ARMATURE VOLTAGE AND THEREBY SPEED. IF THE DRIVE IS FULLY REVERSING, MOTOR ROTATAION CAN ACTUALLY BE REVERSED WITH THIS POT.

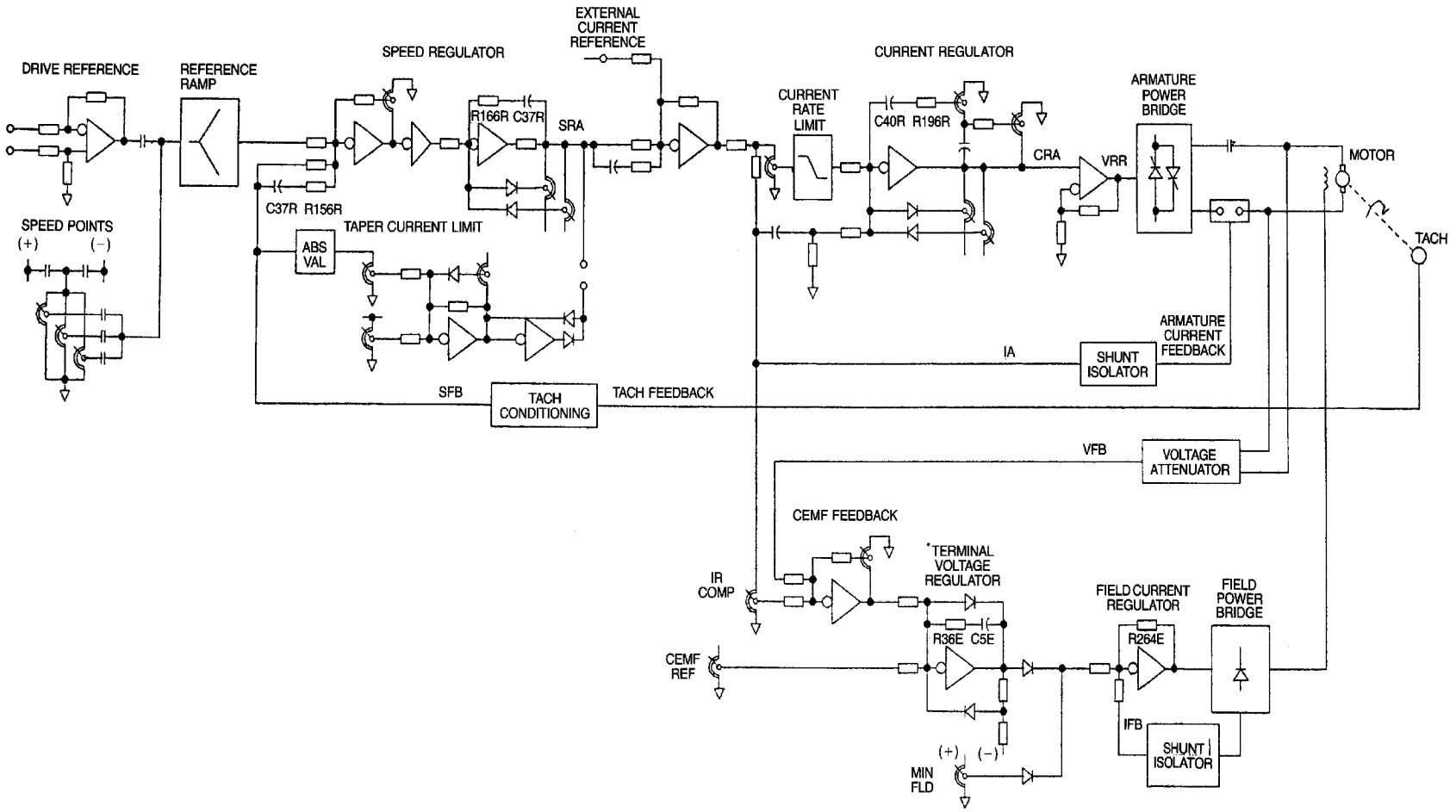
IF THE MOTOR IS RUNNING CORRECTLY NOW, THE PROBLEM IS NOT THE FIRING CIRCUIT, THE BRIDGE, OR THE MOTOR. START TROUBLESHOOTING AT TACHOMETER OUTPUT VOLTAGE, CHECK SPEED FEEDBACK AT TP6, OR REPLACE REGULATOR BOARD.

IF THE MOTOR IS NOT RUNNING CORRECTLY THE PROBLEM COULD BE IN THE FIRING CIRCUIT, GO TO NFCF DIAGNOSTIC ROUTINE SH. 3 OR IT IS A MOTOR PROBLEM.

AFTER THIS TEST IS COMPLETE PUT JUMPER SJ39R, J40R, J41R TO CORRECT POSITIONS AS INDICATED IN THE ELEMENTARY. ALSO DISCONNECT TEST JUMPER TP15 TO TP25.

SILTRON SPEED / CURRENT REGULATOR

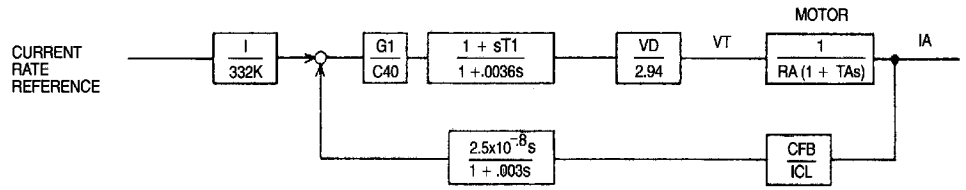
TYPICAL ONE LINE DIAGRAM



*NOTE: IF THE TERMINAL VOLTAGE REGULATOR IS NOT USED, THEN C5 WILL BE SHORTED AND CEMF FDBK DISCONNECTED.

SILTRON CURRENT RATE REGULATOR

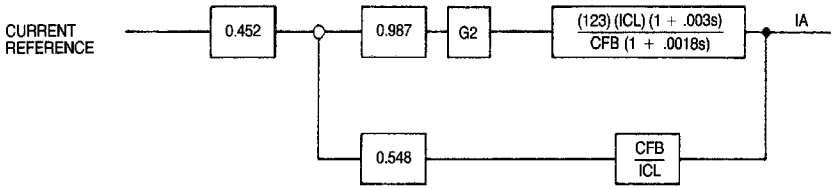
BLOCK DIAGRAM



G1 = CURRENT RATE REGULATOR GAIN (R19) : G1 = 500 WITH R19 AT MIDPOINT
ICL = CURRENT LIMIT (AMPS)
T1 = C40 *R196 = .0067 (SEC.)
CFB = CURRENT FDBK (VOLTS) = 7.5 = I_{MAX}
VD = MOTOR VOLTS (RATED)
NOTE: CURRENT RATE REGULATOR NORMALLY SET AT 500 RAD/SEC.

SILTRON CURRENT REGULATOR

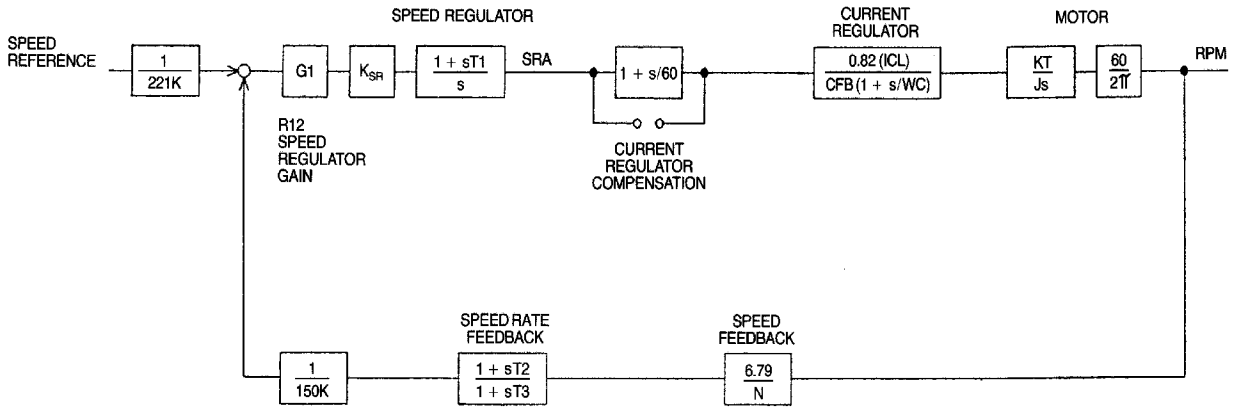
BLOCK DIAGRAM



G2 = CURRENT REGULATOR GAIN, (R25).

SILTRON SPEED / CURRENT REGULATOR BLOCK DIAGRAM

TYPICAL ONE-LINE DIAGRAM



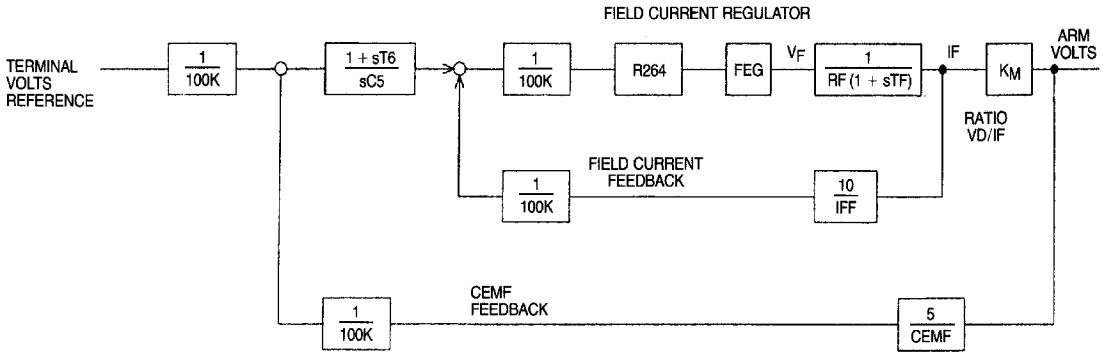
G1 = R12 GAIN; RANGE 1.0 TO 10

$K_{SR} = \frac{(.221)\ (R166)\ K1}{C38}$; K1 = 1.0; J22 – IN
5.5; J21 – IN

T1 = TM = R166 * C38; SPEED REG. CROSSOVER (SEC.)
T2 = T1 = R156*C37 (SEC)
T3 = 0.5*T1 (SEC.)

CFB = CURRENT FDBK (VOLTS)
ICL = CURRENT LIMIT (AMPS)
WC = CURRENT REGULATOR RESPONSE (RAD/SEC.)
N = TOP SPEED (RPM)

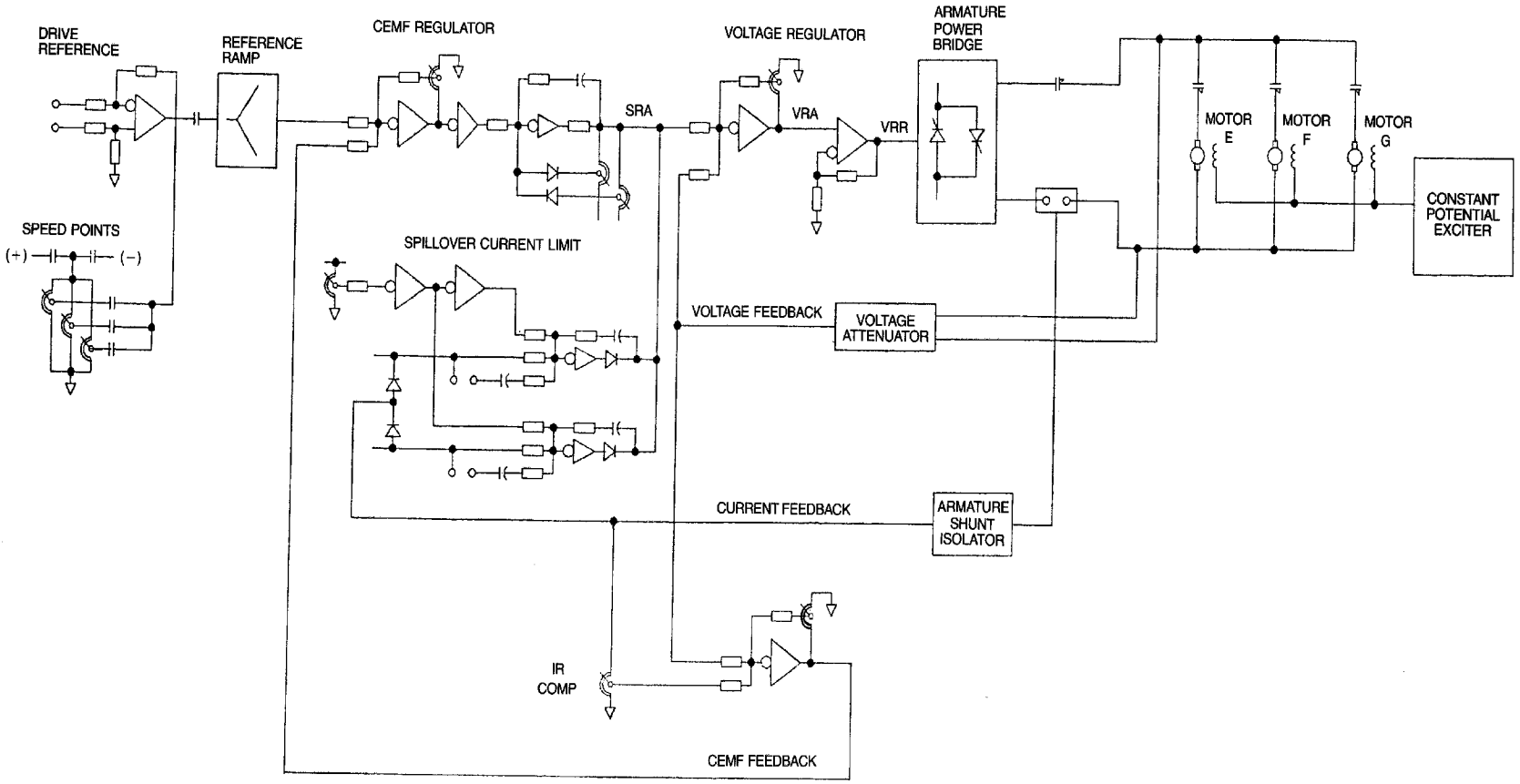
SILTRON TERMINAL VOLTS/FIELD CURRENT REGULATOR



T6 = R36 * C5 (SEC)
FEG = FIELD EXCITER GAIN (TYPICALLY = 15)
IFF = FULL FIELD AMPS
TF = FIELD TIME CONSTANT (SEC)
K_M = RATIO OF ARM. VOLTS TO FIELD AMPS AT WEAK FIELD

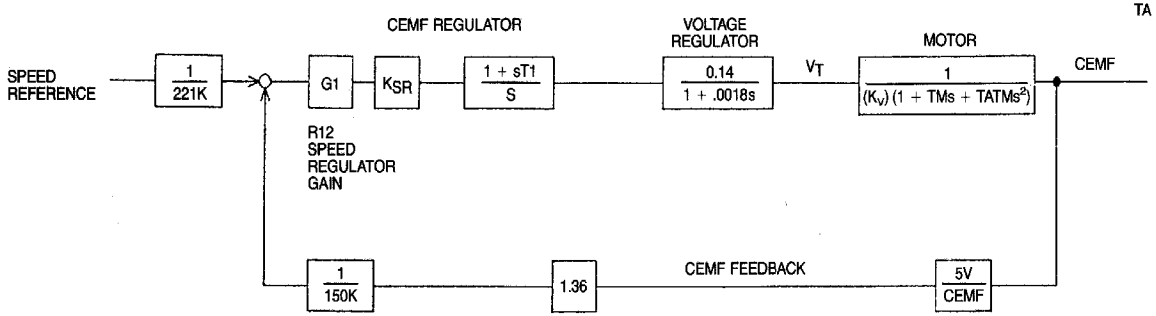
SILTRON CEMF / VOLTAGE REGULATOR

TYPICAL ONE-LINE DIAGRAM



SILTRON CEMF / VOLTAGE REGULATOR

BLOCK DIAGRAM



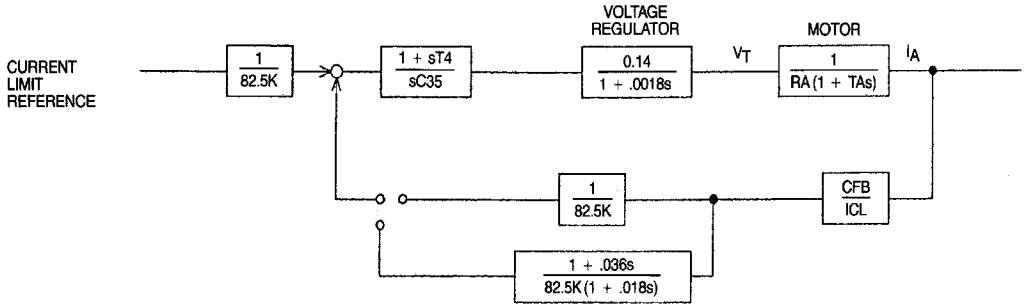
G1 = R12 GAIN; RANGE 1.0 TO 10

$K_{SR} = \frac{(.221)\ (R166)\ K1}{C38}$; K1 = 1.0; J22 – IN
K1 = 5.5; J21 – IN

T1 = TM = R166 * C37; INTERTIA TIME CONSTANT (SEC.)

TA = ARMATURE CIRCUIT TIME CONSTANT (SEC.)

SILTRON CURRENT LIMIT AT STALL



T4 = R151 * C35 (SEC.)

CFB = CURRENT FDBK (VOLTS)

ICL = CURRENT LIMIT (AMPS)

GENERAL ELECTRIC COMPANY • DRIVE SYSTEMS OPERATIONS
SALEM, VIRGINIA 24153

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